The $d8$ accessibility boundary at $Q_4/Q_1 = 5/32$: weight decay pins and sharpens the 1^8 vs. 4+4 transition

frozen-$d4$ representation, 8 bits, full 2^8 cube — 14 accessibility conditions × 10 λ × 3 seeds

Setup

Direct lift of the frozen $d4$ WD experiment (`accessibility_wd_report.pdf`) to $d8$: 8 relevant bits, all $2^8 = 256$ inputs full-batch, two pure-ReLU residual blocks (N=1024, init var $1/N$), same optimizer and convergence conventions. Stage 1 trains embedding + block 1 on the parents $x_1x_2x_3x_4$, $x_5x_6x_7x_8$ with $\lambda = 0$ to convergence, then freezes (checkpoints saved). ReLU homogeneity ($h_1(\alpha) = h_0 + \alpha\,\mathrm{ReLU}(V_1 h_0)$) plus a per-seed **pilot inversion** of the measured $\alpha \mapsto Q_4/Q_1$ map places conditions on 14 target ratios densely bracketing the theory boundary: $\eta_{1^8} = 5Q_1/32\sqrt{8}$ vs. $\eta_{4+4} = Q_4/2\sqrt{2}$ tie at $Q_4/Q_1 = 5/32 = 0.15625$. Stage 2 trains a fresh block 2 + readout on $x_1 \cdots x_8$ with the same λ on both, $\lambda \in \{0, 10^{-5}, \ldots, 10^{-1}\}$, identical stage-2 init across all (ratio, λ) per seed. Circuits are classified from pre-activation Walsh shares over the full 255-mode basis: eight singletons (1^8), four relevant pairs (2^4), two relevant quads (4+4), contribution-weighted over neurons carrying 95% of the $d8$ output. Numerics: training in float32 (float64 is ∼64× slower at 256×1024 on consumer GPUs), all Walsh measurement in float64.

Findings

1. The transition exists and, under decay, sits exactly at 5/32. At $\lambda \geq 10^{-2}$ the $f_{4+4}(Q_4/Q_1)$ curves are a near-step at the theory line (Fig. 3, left): between the two conditions straddling the boundary, f_{4+4} jumps $0.04 \to 0.38$ ($\lambda = 10^{-2}$) and $0.46 \to 0.95$ ($3{\cdot}10^{-2}$) across Q_4/Q_1 : $0.157 \to 0.161$ — a transition localized to ±2% of the predicted value. Left of the line, every λ gives $f_{4+4} \leq 0.05$.

2. Without decay the transition is displaced by dynamics; decay pulls it home. At $\lambda = 0$ the compositional fraction rises only gradually and far to the right ($f_{4+4} = 0.43$ at $Q_4/Q_1 = 0.25$, 0.57 at 0.30): the fast-forming direct circuit partially locks in even when 4+4 is norm-favored. Increasing λ both sharpens the crossover and shifts it monotonically left onto 5/32, saturating by $\lambda \approx 10^{-2}$–$3{\cdot}10^{-2}$; the total stage-2 squared norm collapses $1057 \to 20$ at $Q_4/Q_1 = 0.167$ ($\lambda : 0 \to 10^{-2}$) — the minimum-norm mechanism seen in the $d4$ study, now selecting among three candidate routes. The final MSE itself (Fig. 2) does not see the route change: it is set by λ alone.

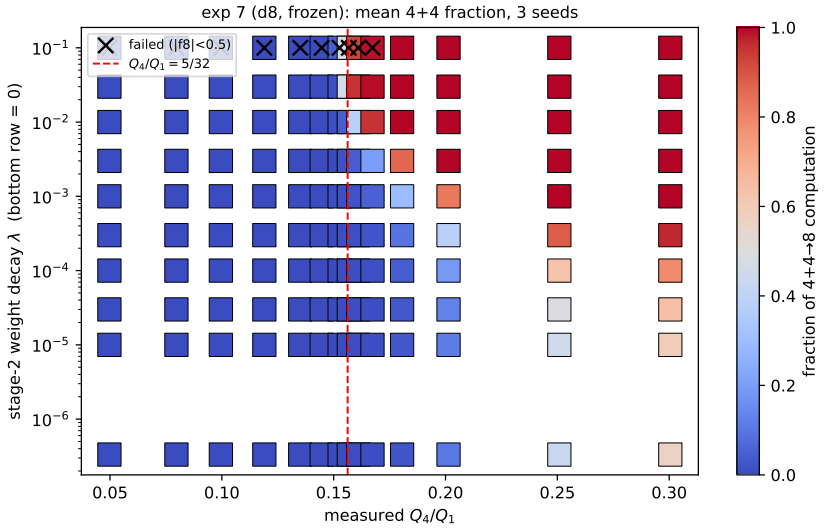


Figure 1: Phase diagram ($Q_4/Q_1, \lambda$), mean 4+4 fraction over 3 seeds; red line: 5/32. Left of the boundary stays direct (blue) at all λ; right of it the compositional phase (red) grows downward as λ decreases the lock-in advantage. × marks $d8$ failure ($|\hat{f}_8| < 0.5$): at $\lambda = 10^{-1}$ every condition with $Q_4/Q_1 \leq 0.168$ fails while all conditions right of the boundary still solve the task through 4+4.

3. There is a WD window where 4+4 survives after 1^8 dies. At $\lambda = 10^{-1}$: $|\hat{f}_8| = 0.37$–0.47 (failed) for every condition with $Q_4/Q_1 \leq 0.168$, but $0.52 \to 0.78$ (alive, rising with Q_4) for $0.18 \to 0.30$, carried by $f_{4+4} = 0.99$ circuits — precisely the ordering $\lambda_c^{4+4} = e_8 Q_4/2\sqrt{2} > \lambda_c^{1^8} = 5e_8 Q_1/32\sqrt{8}$ predicts right of the boundary. The composition ladders (Fig. 4) show the full sequence at $Q_4/Q_1 = 0.167$: pure 1^8 ($\lambda \leq 10^{-3}$) → flip to 4+4 (10^{-2}, with $|\hat{f}_8| \approx 0.95$ intact) →

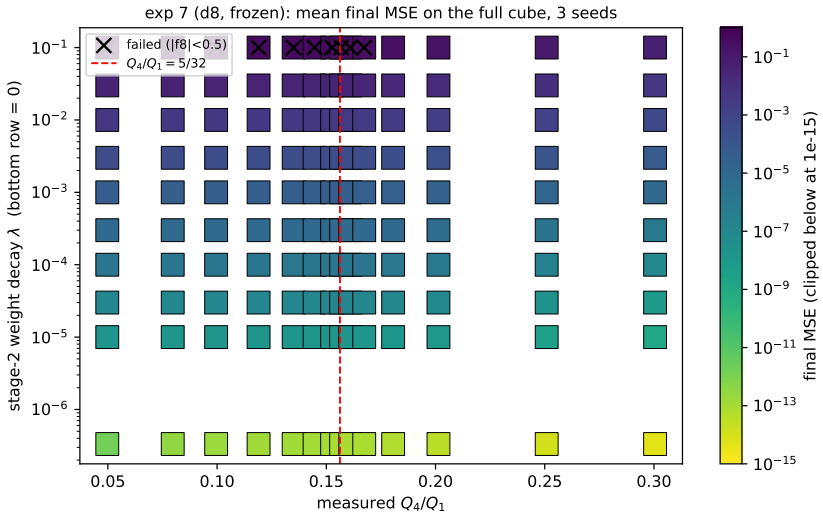


Figure 2: Same cells as Fig. 1, colored by the mean final MSE of the $d8$ output over the full 2^8 cube (log scale, 3 seeds; the full objective is this plus $\lambda\,\|\theta_2\|^2/2$, stored per cell as `wd_penalty`). The MSE is set almost entirely by λ: each row sits near a floor that grows quadratically, MSE $\approx 40\lambda^2$ for $\lambda \geq 10^{-3}$ (from 10^{-13} at $\lambda = 0$ to 0.4 at 10^{-1}), with only a mild decrease to the right (about 10× across the ratio range at every λ). The $1^8 \to 4{+}4$ transition of Fig. 1 leaves no visible step in the MSE; × marks the same failures.

$d8$ suppression (10^{-1}). Left of the boundary (panels 1–2) there is no flip — 1^8 persists until suppression; at 0.25 the flip completes already at $\lambda = 10^{-3}$.

4. The 2^4 route never materializes, as the measured Q_4/Q_2 predicts. The frozen representation gives $Q_4/Q_2 = 0.919$, flat across the whole α-sweep (Fig. 5, right) — close to the ideal-gate prediction $Q_4/Q_2 \approx 1$ and far above the 2^4 viability threshold $1/\sqrt{8} \approx 0.354$. Correspondingly the maximum 2^4 share observed anywhere in the 420 cells is $f_{2^4} = 0.029$. The pilot map (Fig. 5, left) saturates at $Q_4/Q_1 = 0.402$, so all 14 targets were reachable.

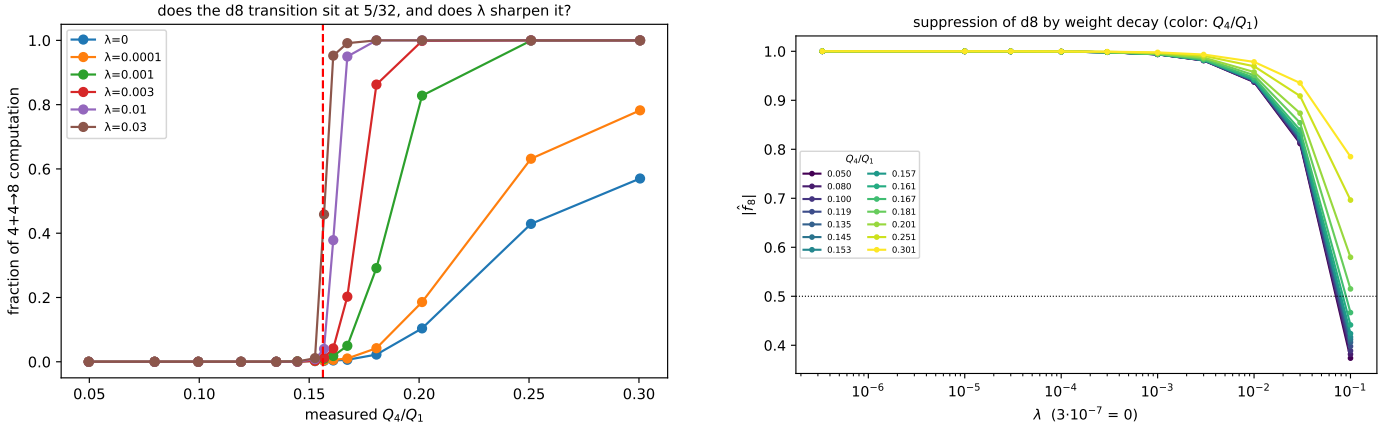


Figure 3: Left: f_{4+4} vs. measured Q_4/Q_1 at fixed λ — increasing decay sharpens the transition and moves it onto 5/32 (red). Right: $|\hat{f}_8|$ vs. λ per condition — suppression is ordered by accessibility; the 0.5 failure line (dotted) is crossed first by the low-Q_4 conditions.

Answers to the five questions. (i) Yes: under meaningful decay the $1^8 \leftrightarrow 4{+}4$ transition sits at $Q_4/Q_1 = 5/32$ to within the grid spacing (±2%). (ii) λ both sharpens the transition and removes the dynamical bias that displaces it rightward at $\lambda = 0$. (iii) Yes: at $\lambda = 10^{-1}$ the direct route is dead everywhere while 4+4 still learns $d8$ for $Q_4/Q_1 \geq 0.18$. (iv) $d8$ learning fails at $\lambda \sim 10^{-1}$ left of the boundary (and begins eroding from $3{\cdot}10^{-2}$), later the larger Q_4 — the $\lambda_c = e_8\eta$ ordering. (v) $Q_4/Q_2 = 0.92 \approx 1$ as the ideal 1^4-gate spectrum predicts, and 2^4 is correspondingly absent (≤ 0.03 everywhere).

Artifacts: `gdrive:parity/analysis_output/accessibility_d8_wd/` (`results_merged.csv`, 6 worker slices, trajectories, pilots, figures, this report); stage-1 checkpoints in `accessibility_ckpts/d8pre_seed{0,1,2}.pt`; code: `analysis/accessibility/run_d8_wd_experiment.py`, `merge_d8_wd.py`. Companion: `accessibility_wd_report.pdf` ($d4$).

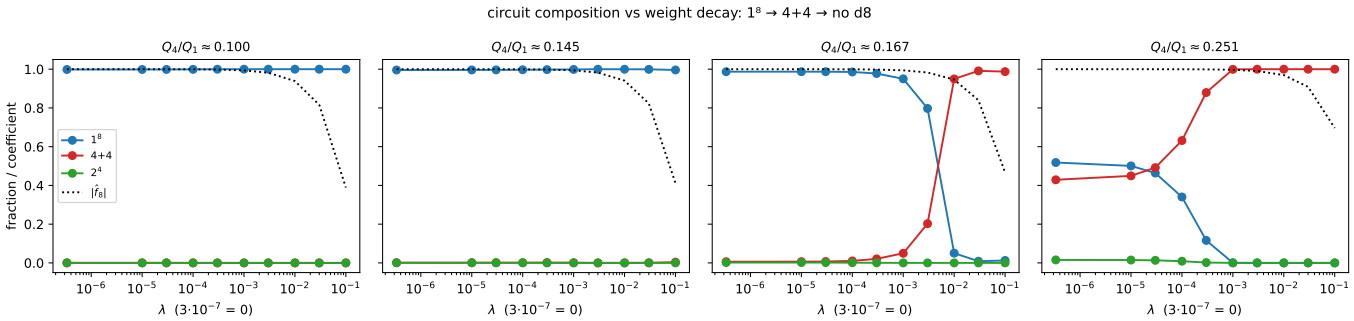


Figure 4: Circuit composition vs. λ at four representative accessibilities. The predicted ladder $1^8 \to 4{+}4 \to$ no-$d8$ appears exactly and only right of the boundary; 2^4 (green) stays at zero everywhere.

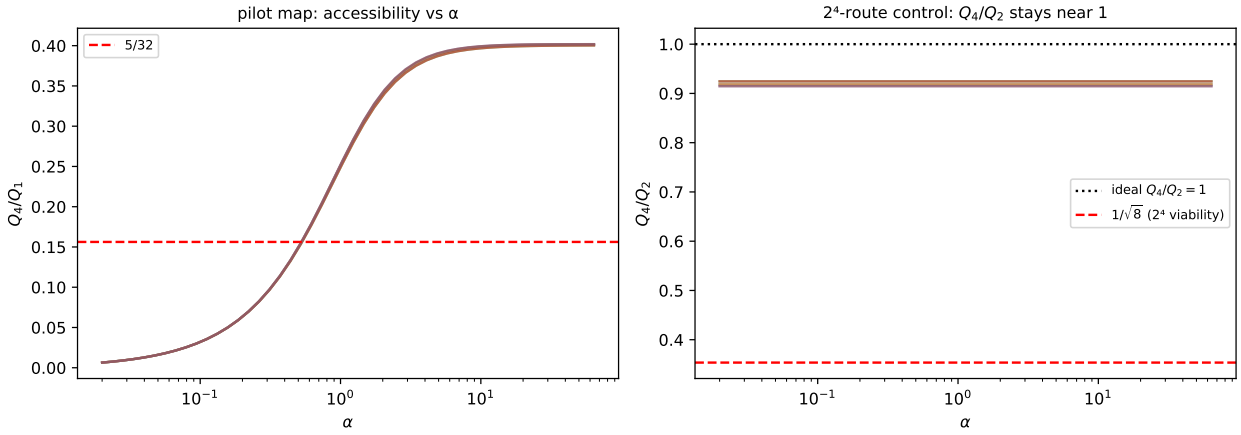


Figure 5: Controls. Left: pilot map $\alpha \mapsto Q_4/Q_1$ (all seeds overlap), crossing 5/32 near $\alpha \approx 0.55$ and saturating at 0.402. Right: $Q_4/Q_2 = 0.919$ throughout — near the ideal 1^4-gate prediction of 1 and far above the 2^4 viability line $1/\sqrt{8}$, explaining the absent 2^4 route.